The complete high-temperature Ising susceptibility: a constrained-coordinate proof of the natural boundary

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Abstract

We prove that the complete zero-field isotropic square-lattice Ising susceptibility, continued from high temperature, has the unit circle as a natural boundary in $s = \sinh(2\beta_{\mathrm{phys}}J)$. An elementary bordered determinant represents the odd-particle pair product with a constant final row and column. On the exact full-lattice angular constraint a balanced deformation preserves the total angle. A regular coordinate and one collective tangential field then give a logarithmic absolute bound for the entire differentiated higher odd-particle sum. The same estimate controls finite-order complements. At a dense family of odd prime roots, two reflected collision charts give a nonzero inverse-distance derivative, evaluated by a rational radial primitive. Signed constraints treat the two boundary halves, with the physical normalization and the origin included throughout.

1 The physical function and the proof

The high-temperature susceptibility of the square-lattice Ising model is represented by an infinite series of odd-particle integrals. The singularities of individual terms accumulate densely on the unit circle, and their survival in the complete sum is the central analytic issue. Orrick, Nickel, Guttmann, and Perk [1, Section 3.2] and Boukraa et al. [2] studied the singularities and the series evidence for a natural boundary. We use a balanced deformation of the constrained angular integral, a regular inverse coordinate, and a single tangential field to estimate the full differentiated tail.

Let $J > 0$ and consider the zero-field isotropic Ising model on the infinite square lattice. For $0 < s = \sinh(2\beta_{\mathrm{phys}}J) < 1$, let

$$\mathcal{X}_H(s) = \beta_{\mathrm{phys}}^{-1}\chi_H(s) = \sum_{(m,n)\in\mathbb{Z}^2} \langle \sigma_{00}\sigma_{mn}\rangle_s.$$

The high-temperature infinite-volume Gibbs state has zero magnetization. This sum includes every lattice site and its origin. We use $t = s^{-1}$ only as an exterior coordinate and put

$$S(t) = t + t^{-1}, \qquad Q_H(t) = t(1 - t^{-4})^{1/4},$$

where the fourth root tends to one at infinity.

Theorem 1.1. *The physical high-temperature germ of χ_H admits no local holomorphic continuation across any point of $|s| = 1$. The same is true of the normalized complete susceptibility $\mathcal{X}_H$.*

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The real-temperature physical inputs are the odd form-factor expansion of the full correlation and its full-lattice angular sum [1, eqs. (1.9), (1.11), (3.13), (3.17)–(3.20), (4.1)–(4.2)], also recorded in [2, eqs. (3), (5)–(9)]. Their use is restricted to the physical real interval. We establish complex continuation and the boundary estimates below. For the complete angular coefficients C_N we prove

$$\mathcal{X}_H(1/t) = Q_H(t) \sum_{\substack{N\geq 1\\ N \text{ odd}}} C_N(t).$$

At $t_* = e^{\mathrm{i}\alpha}$, $\alpha = 2\pi a/p \in (0,\pi)$, for each odd prime $p \geq 5$, set $k = (p^2-1)/2$. The decisive estimates are

$$C_p^{(k)}((1+\varepsilon)t_*) = \Lambda_\alpha \varepsilon^{-1} + o(\varepsilon^{-1}), \qquad \Lambda_\alpha \neq 0, \qquad \sum_{\substack{N>p\\ N \text{ odd}}} |C_N^{(k)}((1+\varepsilon)t_*)| = O(1+|\log\varepsilon|).$$

The needed fixed derivatives of lower particle orders are logarithmically bounded. Thus the complete infinite sum cannot be holomorphic at these dense points. Dividing by the locally nonzero physical factors transfers the obstruction to χ_H.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

2 The exact full-lattice angular representation

Write

$$W(y;t) = S(t) - \tfrac{1}{2}(y+y^{-1}), \qquad z = W - \sqrt{W^2-1}, \qquad \sinh\gamma = \tfrac{1}{2}(z^{-1}-z), \quad z = e^{-\gamma}.$$

For $|t|>1$ and $|y|=1$ choose the unique root with $|z|<1$, continued from the positive real interval. Indeed $S(t) \notin [-2,2]$, so W cannot lie in $[-1,1]$. The root and $\sinh\gamma$ are single-valued there. For $y_i = e^{\mathrm{i}\theta_i}$ define the complete pair

$$P_{ij} = \left[\frac{\sin((\theta_i-\theta_j)/2)}{\sinh((\gamma_i+\gamma_j)/2)}\right]^2 = -\frac{(y_i-y_j)^2 z_i z_j}{y_i y_j (1-z_i z_j)^2}. \tag{2.1}$$

Only the rational square is used globally, so logarithmic and half-angle sign choices do not affect the integral. Let $Z = \prod_i z_i$, and for $\sigma \in \{1,-1\}$ set $a_+ = 0$, $a_- = \pi$, and

$$\mathbb{H}_N^\sigma = \{\theta \in \mathbb{T}^N : \textstyle\sum_i \theta_i = a_\sigma \pmod{2\pi}\}, \qquad d\mu_N = \textstyle\prod_{i<N} \frac{\mathrm{d}\theta_i}{2\pi}.$$

$$\theta_N = a_\sigma - \sum_{i<N} \theta_i \pmod{2\pi}.$$

This is normalized Haar measure; its Euclidean hypersurface normalization is $[\sqrt{N}(2\pi)^{N-1}]^{-1}$. For odd N define

$$C_N^\sigma(t) = \frac{1}{N!}\int_{\mathbb{H}_N^\sigma} \frac{1+\sigma Z}{1-\sigma Z} \prod_i \frac{1}{\sinh\gamma_i} \prod_{i<j} P_{ij}\, d\mu_N, \qquad C_N = C_N^+. \tag{2.2}$$

The minus sign is an auxiliary angular constraint, not a different physical observable in the theorem.

Proposition 2.1 (Physical identification and normal convergence)**.** *The complete normalized physical susceptibility has the exterior continuation*

$$\mathcal{X}_H(1/t) = Q_H(t) \sum_{\substack{N\geq 1\\ N \text{ odd}}} C_N(t). \tag{2.3}$$

Both signed series, with each fixed number of derivatives, converge normally on compact subsets of $|t| > 1$. The origin contributes exactly one to $\mathcal{X}_H$.

Proof. For real $t > 1$, the per-site physical input is

$$\langle \sigma_{00}\sigma_{mn}\rangle_{1/t} = Q_H(t) \sum_{\substack{N\geq 1\\ N \text{ odd}}} c_N(m,n;t),$$

$$c_N(m,n;t) = \frac{1}{N!}\int_{\mathbb{T}^N} Z^{|m|} e^{\mathrm{i}n\sum\theta_i} \prod_{i<j} P_{ij} \prod_i \frac{\mathrm{d}\theta_i}{2\pi \sinh\gamma_i}.$$

The exponential is equivalent under angle reversal to the cosine in [1, eq. (4.2)]. In the isotropic specialization $\cot\alpha_{\text{anis}} = 1$, its $k = s^2 = t^{-2}$ and $k^{-1/2}(1-k^2)^{1/4} = Q_H(t)$; the symbols α_{anis} and k here belong only to that source normalization. The dispersion and pair factors agree literally with (2.1).

The bordered determinant identity proved in Section 4 has size $N+1$ with bounded entries on a common small complex angular strip. There the one-body factors are bounded and $|z_i| \leq e^{-c} < 1$. Shifting every angle by $\mathrm{i}\eta\,\mathrm{sgn}(n)$, for fixed sufficiently small $\eta > 0$, gives

$$|c_N(m,n;t)| \leq \frac{C^N N^{(N+1)/2}}{N!} e^{-N(c|m|+\eta|n|)}.$$

Periodicity justifies the shift; Hadamard supplies the determinant bound. This is summable in m, n and all odd N. Thus particle and lattice sums may be interchanged. The m-sum gives $(1+Z)/(1-Z)$. For a periodic analytic F on $\mathbb{T}^N$, define

$$A(v) = \int_{\mathbb{T}^{N-1}} F(\theta_1,\dots,\theta_{N-1}, v - \textstyle\sum_{i<N}\theta_i) \prod_{i<N} \frac{\mathrm{d}\theta_i}{2\pi}.$$

Its Fourier coefficient of index $-n$ is the N-angle integral with multiplier $e^{\mathrm{i}n\sum\theta_i}$. The absolutely convergent Fourier sum is $A(0)$, giving precisely (2.2) with the normalized Haar measure $d\mu_N$. At the origin the per-site sum is $\langle\sigma_{00}^2\rangle = 1$, so the origin is already contained in this full-lattice sum. For a further normalization check,

$$C_1(t) = \frac{1}{S(t)-2} = \frac{s}{(1-s)^2}.$$

On a compact exterior neighborhood and a slightly larger one, $m \leq |z_i| \leq q < 1$ uniformly on the angular torus. All one-body and bordered entries are bounded, and the signed product denominator has a fixed gap. Since the Haar measure is one,

$$|C_N^\sigma(t)| \leq C_0\sqrt{N}(C_1/\sqrt{N})^N.$$

The bound is summable; Cauchy's estimate on the larger compact neighborhood proves normal convergence of every fixed derivative. The identity theorem gives (2.3) on the exterior. Near $s = 0$, the interior root has the form $z = s\,a(s,y)$ with analytic bounded a, and $1/\sinh\gamma = s\,b(s,y)$ with analytic bounded b. The same determinant estimate gives normal convergence of $\sum C_N(1/s)/s$. Multiplication by $(1-s^4)^{1/4}$ therefore fills the removable point $s = 0$ and recovers the full high-temperature analytic germ. □

The signed coefficients commute with complex conjugation and obey, for odd N,

$$C_N^+(-t) = -C_N^-(t). \tag{2.4}$$

Indeed replace y_i, z_i, t by $-y_i, -z_i, -t$. The angular constraint becomes $\prod y_i = -1$, every pair is unchanged, all N one-body factors change sign, and Z changes sign. Angular translation preserves Haar measure. In particular no evenness of the physical high-temperature susceptibility is asserted.

3 An equal-angle odd-prime family

Lemma 3.1. *Let $p \geq 5$ be prime, $1 \leq a \leq (p-1)/2$, and put*

$$\alpha = 2\pi a/p, \quad t_* = e^{\mathrm{i}\alpha}, \quad S_* = 2\cos\alpha, \quad N_* = p, \quad k = (p^2-1)/2.$$

These points are dense in both open upper quadrants, $S_* \neq 0$, and $k \equiv 0 \pmod 4$. The first odd order admitting $S_* = \cos(2\pi l/n) + \cos(2\pi m/n)$ is p. At that order the unordered cosine pair is uniquely $\{\cos\alpha, \cos\alpha\}$. For $b = \arccos(|S_*| - 1)$, the algebraic number $e^{-\mathrm{i}b}$ is not a root of unity.

Proof. Write $d = (p-1)/2$, $\zeta_p = e^{2\pi\mathrm{i}/p}$, $u_j = \zeta_p^j + \zeta_p^{-j}$, and $E = \mathbb{Q}(\zeta_p + \zeta_p^{-1})$. The $u_1, \ldots, u_d$ are linearly independent over $\mathbb{Q}$: a relation gives a polynomial of degree at most $p-1$, with zero constant term, vanishing at ζ_p; it is a multiple of $1 + X + \cdots + X^{p-1}$, and that multiple must be zero. The conjugates of each u_a are exactly $u_1, \ldots, u_d$, with $u_1 > 0$ and $u_d < 0$. Also u_a is irrational: otherwise ζ_p^a would satisfy $X^2 - u_a X + 1$ over $\mathbb{Q}$, contrary to its degree $p-1 > 2$. If $p \nmid n$, the cyclotomic fields $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_n)$ have intersection $\mathbb{Q}$: their compositum has degree $\varphi(pn) = \varphi(p)\varphi(n)$. A cosine representation at an odd $n < p$ would contradict irrationality. Order p realizes the equal pair.

Put $\delta = \operatorname{sgn}(u_a)$ and $S_+ = \delta u_a = |S_*|$. The number S_+ has a negative conjugate, using u_d when $\delta = 1$ and u_1 when $\delta = -1$. If $\xi = e^{-\mathrm{i}\arccos(S_+ - 1)}$ were a root of unity, every conjugate of $\xi + \xi^{-1} = 2S_+ - 2$ would lie in $[-2, 2]$. The indicated conjugate is strictly less than -2, a contradiction. Here each embedding of $\mathbb{Q}(S_+)$ extends to the field containing ξ. This proves the asserted endpoint nonresonance for both signs at once.

At odd order p the endpoint cosine -1 is absent. A cosine 1 has an impossible partner $S_* - 1 < -1$ when $S_* < 0$; for $S_* > 0$ it is excluded by the preceding endpoint argument. All remaining twice-cosines are u_j. The equation $2u_a = u_l + u_m$ and independence force $l = m = a$ modulo reflection. The only angular equality centers are therefore $\theta_i = -\alpha$ and $\theta_i = +\alpha$ for every i. Approximation of any $\alpha_0 \in (0, \pi)$ by nearest prime-grid angles proves density. Finally every odd square is 1 modulo 8. □

For a point with negative S_*, let $\widehat{t_*} = -\overline{t_*} = e^{\mathrm{i}(\pi - \alpha)}$ and use C_N^- there. Equation (2.4) gives

$$C_N^{(j)}((1+\varepsilon)t_*) = (-1)^{j+1}\overline{(C_N^-)^{(j)}((1+\varepsilon)\widehat{t_*})}. \tag{3.1}$$

A signed equality at odd order n has both angles in $(2\pi\mathbb{Z} + \pi)/n$. Adding π to both angles converts it to an ordinary odd-order equality at $-S_*$. Thus the same arithmetic proves uniqueness and absence of lower orders on the signed half. Nonresonance gives $e^{-\mathrm{i}nb} \neq \pm 1$ for every integer $n \geq 1$.

4 A bordered determinant for odd particles

The physical pair product has a determinant with a constant border. This keeps both its collision zeros and its factorial estimates visible.

Lemma 4.1 (Four points on the dispersion curve)**.** *Let nonzero y_i, z_i satisfy $y_i+y_i^{-1}+z_i+z_i^{-1} = 2S$ for $1 \le i \le 4$, and set $B_{ij} = (y_i - y_j)/(1 - z_i z_j)$. Then*

$$B_{12}B_{34} - B_{13}B_{24} + B_{14}B_{23} = \left(\prod_{i=1}^{4} \frac{z_i}{y_i}\right) \prod_{i<j} B_{ij}.$$

The equality holds wherever the expressions are defined and, after clearing denominators, at every point of the curve.

Proof. Write $a_{ij} = y_i - y_j$, $t_i = z_i + z_i^{-1}$, and $Z_4 = z_1 z_2 z_3 z_4$. For a partition $ij \mid kl$ of the four indices, direct multiplication gives

$$\frac{\prod_{v\in\{i,j\},\, w\in\{k,l\}}(1 - z_v z_w)}{Z_4} = R + t_i t_j + t_k t_l, \qquad R = Z_4 + Z_4^{-1} + 2 - \sum_{v<w} (z_v z_w + (z_v z_w)^{-1}).$$

Multiplying the proposed identity by $\prod_{i<j}(1 - z_i z_j)$, the common R term cancels because $a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$. The remaining left side, divided by Z_4, is

$$\begin{aligned} a_{12}a_{34}(t_1t_2 + t_3t_4) - a_{13}a_{24}(t_1t_3 + t_2t_4) \\ + a_{14}a_{23}(t_1t_4 + t_2t_3) = \det[\,1,\ y_i,\ t_i,\ y_i t_i\,]_{i=1}^{4}. \end{aligned}$$

Here the last equality is a two-column Laplace expansion. Since $t_i = 2S - y_i - y_i^{-1}$, column operations give

$$\det[1, y_i, t_i, y_i t_i] = \det[1, y_i, y_i^{-1}, y_i^2] = \frac{\det[y_i, y_i^2, 1, y_i^3]}{\prod_i y_i} = \frac{\prod_{i<j}(y_i - y_j)}{\prod_i y_i}.$$

This is the required cleared numerator. □

Lemma 4.2 (Pfaffian propagation)**.** *Suppose a skew matrix K has the property that every four-index principal submatrix satisfies*

$$K_{ab}K_{cd} - K_{ac}K_{bd} + K_{ad}K_{bc} = \prod_{v<w\in\{a,b,c,d\}} K_{vw} \quad (a < b < c < d).$$

Then, at every even order for which the intervening products can be canceled, $\operatorname{Pf} K = \prod_{i<j} K_{ij}$. *For a rational or analytic family with a dense set of such points, the identity extends to the whole family.*

Proof. Let I be the first $N-4$ indices and let a, b, c, d be the last four. The elementary Pfaffian condensation identity is

$$\operatorname{Pf} K_{Iabcd} \operatorname{Pf} K_I = \operatorname{Pf} K_{Iab} \operatorname{Pf} K_{Icd} - \operatorname{Pf} K_{Iac} \operatorname{Pf} K_{Ibd} + \operatorname{Pf} K_{Iad} \operatorname{Pf} K_{Ibc}.$$

For completeness, when $K_I = A$ is invertible, the simultaneous row and column elimination in $\left[\begin{smallmatrix} A & B \\ -B^T & C \end{smallmatrix}\right]$ has determinant one and gives the diagonal blocks A and $C + B^T A^{-1} B$. Pfaffian multilinearity therefore reduces the displayed formula to the defining four-by-four Pfaffian formula. The identity is polynomial, so it holds also when A is singular.

Assume the product formula at all smaller even orders. Writing $P_I = \prod_{i<j\in I} K_{ij}$, the right side factors as

$$P_I^2 \prod_{i\in I,\, v\in\{a,b,c,d\}} K_{iv}(K_{ab}K_{cd} - K_{ac}K_{bd} + K_{ad}K_{bc}) = P_I \prod_{i<j} K_{ij}.$$

The left side is $P_I \operatorname{Pf} K$. Cancel P_I. The orders two and four start the induction. □

Proposition 4.3 (Odd bordered identity)**.** *For odd N, let $\mathcal{A}$ be the skew $(N+1)$-by-$(N+1)$ matrix*

$$\mathcal{A}_{ij} = L_{ij} := -\mathrm{i}\frac{y_i - y_j}{1 - z_i z_j} \quad (i,j \le N), \qquad \mathcal{A}_{i,N+1} = 1, \quad \mathcal{A}_{N+1,i} = -1.$$

On the dispersion curve $y_i + y_i^{-1} + z_i + z_i^{-1} = 2S$,

$$\det \mathcal{A} = \prod_{i<j} P_{ij}. \tag{4.1}$$

The identity is rational and has no square-root choices.

Proof. Choose local $r_i^2 = z_i/y_i$ and put $K_{ij} = -\mathrm{i} r_i r_j (y_i - y_j)/(1 - z_i z_j)$. The four-point lemma gives $\operatorname{Pf} K_{\{1,2,3,4\}} = \prod_{i<j\le 4} K_{ij}$: both sides equal $-(\prod_i r_i^3)\prod_{i<j} B_{ij}$. Pfaffian propagation proves this product identity at every even order, first at generic points and then by continuation.

Append one point $y_0 = R$ on the curve, using the branch $z_0 = -R^{-1} + O(R^{-2})$ and $r_0 = -\mathrm{i}R^{-1} + O(R^{-2})$ as $R \to \infty$. Then $K_{i0} \to r_i$. Squaring the even identity at order $N+1$ gives

$$\det \begin{pmatrix} K & r \\ -r^T & 0 \end{pmatrix} = \left(\prod_i r_i^2\right) \prod_{i<j} K_{ij}^2.$$

Apply the congruence $A \mapsto DAD$ with $\operatorname{diag}(r_1^{-1}, \dots, r_N^{-1}, 1)$. Its determinant is multiplied by $\prod_i r_i^{-2}$, and the resulting matrix is exactly $\mathcal{A}$. Since $K_{ij}^2 = P_{ij}$, this proves (4.1). The denominators may be cleared to extend across removable cases. For $N = 1$, the matrix is $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ and the identity reads $1 = 1$. □

Lemma 4.4 (Minors and collisions)**.** *If the ordinary entries L_{ij} are bounded by a, a determinant with ℓ marked ordinary entries, including any fixed derivatives of them, has its unmarked factor bounded by*

$$C_j N^{C_j} (C\sqrt{N}\, a)^{N-1-\ell}$$

when the marked entries and their fixed jets are bounded. Here $\ell \le j$ and j is fixed; terms with more marked entries than the polynomial permits vanish. On a regular fixed-order collision chart, $\det \mathcal{A}$ *has the complete squared Vandermonde factor of degree $N(N-1)$.*

Proof. A differentiated determinant expands into polynomially many choices of marked rows and columns, leaving complementary minors. The ghost row and column are never differentiated because their entries are constant. Expand both in each remaining minor. Each nonzero term contains exactly two border entries and a minor of the ordinary block of size $N - 1 - \ell$. There are at most N^2 choices; Hadamard bounds that minor. Fixed repeated derivatives of a marked entry do not remove additional rows or columns, and their jets have the stated bound. For the collision assertion use (4.1): each rational pair has its double zero at $y_i = y_j$. Permutation covariance of the coupled contour gives the same zero at $\theta_i = \theta_j$. On a compact regular collision chart its analytic or smooth quotient has bounded fixed jets. □

5 A product-preserving deformation of the whole-bulk integral

Fix either signed family from Section 3, a positive-geometry point t_* with $S_* = t_* + t_*^{-1} \in (0,2)$, and its sign σ. Write $\mathcal{C}_N = C_N^\sigma$, $N_* = p$, $\widehat{\alpha} = \arccos(S_*/2) \in (0, \pi/2)$, and

$$t_\varepsilon = (1+\varepsilon)t_*, \qquad b = \arccos(S_* - 1), \qquad \xi = e^{-ib}, \qquad H = \log(1/\varepsilon).$$

The constants in this section may depend on the selected point and a fixed derivative order k, but never on the odd particle number N or on ε. The infinite-tail estimates concern $N \ge N_* + 2$,

where $k = (N_*^2 - 1)/2$. We prove logarithmic absolute bounds through k, which are smaller than the first singular term. The deformation and the regular transport formulas are also used at fixed odd orders $3 \le N \le N_*$ in Section 10.

For the regular dispersion coordinate we write

$$\varphi = \arccos W, \qquad z = e^{-i\varphi}, \qquad \gamma = i\varphi.$$

Its physical continuation lies in the fourth quadrant near either inverse-dispersion branch. The angular integration coordinates remain θ_i, with $\sum_i \theta_i = a_\sigma \pmod{2\pi}$.

Choose a smooth periodic function p supported in $(0, \pi)$, with $0 \le p \le 1$ and $p = 1$ on $[\delta, \pi - \delta]$. Here $\delta > 0$ is smaller than $\widehat{\alpha}/2$, $b/2$, and $(2\widehat{\alpha} - b)/2$, and is small enough that b is inside this plateau and $1 + g'(\theta) > 0$ on the upper endpoint sliver. Here $g(\theta) = \arccos(S_* - \cos\theta)$; the choices are possible because $b < 2\widehat{\alpha}$ and $g'(0) = 0$. Put $m(\theta) = p(-\theta)$. Thus p, m have disjoint supports, and both inverse branches have fixed plateau neighborhoods. Define

$$\begin{aligned} P &= \sum_i p(\theta_i), \qquad Q = \sum_i m(\theta_i), \\ h_i &= \frac{\tau}{N}\{Pm(\theta_i) - Qp(\theta_i)\}, \qquad y_i(\lambda) = e^{i\theta_i + \lambda h_i}, \quad 0 \le \lambda \le 1. \end{aligned} \tag{5.1}$$

The positive number τ is fixed and sufficiently small. In particular $\sum_i h_i = 0$, so $\prod_i y_i = \sigma$ throughout the deformation.

Lemma 5.1 (Exact constrained deformation)**.** *The full integral defining $\mathcal{C}_N$ is unchanged by (5.1). Its angular Jacobian and its fixed jets have bounds $C_j^N N^{C_j}$, uniformly in N.*

Proof. For $y = e^{v + i\theta}$,

$$\Im W = \Im S - \sinh v \sin\theta.$$

On the upper semicircle the motion in (5.1) is inward, and on the lower semicircle it is outward. Both motions increase $\Im W$. For the admissible exterior parameters it remains positive, so every z_i stays on the interior sheet. Thus $1 - \sigma Z$ and all the pair denominators $1 - z_i z_l$ stay nonzero. The kernel square (2.1) has no $1 - y_i y_l$ denominator. The closed constrained torus therefore undergoes a legal holomorphic homotopy.

Write $w_i = \theta_i - i h_i$, so $\sum_i w_i = \sum_i \theta_i$. The full matrix $I - iDh$ preserves the tangent hyperplane and induces the identity on its one-dimensional quotient. Its full determinant is consequently its constrained tangent determinant. Directly,

$$Dh = \frac{\tau}{N} \operatorname{diag}(Pm_i' - Qp_i') + \frac{\tau}{N}(m \otimes p' - p \otimes m').$$

The diagonal and the two rank-one terms have operator norms at most $C\tau$, independently of N. Hence the Jacobian has modulus at most C^N, and its inverse matrix is uniformly bounded after reducing τ. Differentiating the determinant, or its logarithmic derivative formula, shows that fixed jets add only a polynomial in N. □

Let η be smooth, $0 \le \eta \le 1$, with $\eta(\mu) = 0$ for $\mu \le 1/4$ and $\eta(\mu) = 1$ for $\mu \ge 1/2$. Partition the final deformed integral with the three weights

$$w_F = \eta(P)\eta(Q), \qquad w_L = \tfrac{1}{2}(1 - \eta(P))(1 + \eta(Q)), \qquad w_U = \tfrac{1}{2}(1 - \eta(Q))(1 + \eta(P)).$$

Their sum is one. Reflection $\theta \mapsto -\theta$ exchanges P, Q, takes y_i to y_i^{-1}, and preserves z_i. Every rational pair in (2.1) is unchanged; so is the bordered determinant by (4.1). The constraint $\prod y_i = \sigma$ and Haar measure are preserved for either sign. Therefore

$$\mathcal{C}_N = F_N + 2L_N, \tag{5.2}$$

where F_N and L_N are the final integrals with weights w_F and w_L. All weights are imposed after the full closed homotopy. The support of L_N has $P \le 1/2$; it excludes the upper inverse branch and consists of the lower semicircle together with two small upper endpoint slivers.

5.1 The sector containing both branch populations

Lemma 5.2. *For every fixed $j \geq 0$ and every odd $N \geq 5$,*

$$|F_N^{(j)}(t_\varepsilon)| \leq C_j N^{3j+2} \left(\frac{C_j}{\sqrt{N}}\right)^N.$$

In particular these derivatives have a uniformly bounded absolute sum.

Proof. On this support $P, Q \geq 1/4$. Both bump supports stay a fixed distance from the endpoints $0, \pi$. The imaginary-part formula, and the bounded range of W, imply

$$-\log|Z| \geq c\left(N\varepsilon + \frac{\tau PQ}{N}\right) \geq \frac{c}{N}.$$

Every true lower branch has outward displacement $\tau P/N$, and every true upper branch has inward displacement $\tau Q/N$. Their normal forms give a branch distance at least c/N in the W plane. All other coordinates stay in fixed regular neighborhoods. Consequently $|\partial_t \log z_i| \leq C\sqrt{N}$ on a disk $|t - t_\varepsilon| \leq c/N^3$. Reduce c once: the variation of $\log Z$ is at most $CN^{3/2}c/N^3$, smaller than half its preceding gap. The true branches remain on their interior sheets throughout this disk.

We estimate the bordered determinant directly. Entries with both points near the same inverse branch are bounded: the common factor $\varphi_i + \varphi_l$ cancels, leaving $(\varphi_i - \varphi_l)$ times a bounded analytic coefficient. Away from an opposite-branch pair, compactness gives the same bound. Let B be the number of coordinates in the two fixed true-branch neighborhoods. If a pair is near opposite branches, its numerator is bounded and

$$|1 - z_i z_l| \geq c\sqrt{(P+Q)/N}, \qquad |L_{il}| \leq C\sqrt{N/(P+Q)} \leq C\sqrt{N/B}.$$

Indeed both regular branch coordinates remain in a common sector of width strictly less than π (a small fixed enlargement of the fourth quadrant suffices), while their imaginary dispersion margins are at least $c\tau P/N$ and $c\tau Q/N$. Each such coordinate contributes one to $P + Q$. There are at most B exceptional entries in any row; all remaining entries are bounded. Every ordinary row therefore has Euclidean norm at most $C\sqrt{N}$; the constant border row has norm $\sqrt{N}$ as well. Hadamard gives

$$|\det \mathcal{A}| \leq C^N N^{(N+1)/2}$$

uniformly on the disk.

The one-body factors have a common independent angular majorant

$$f(\theta) = C\left(1 + \operatorname{dist}(\theta, \{b, -b\})^{-1/2}\right).$$

To verify this, use the imaginary part of the branch normal form when $|u| \leq C(\varepsilon + |h|)$ and its linear real part outside that interval. The small parameter disk is absorbed in the positive branch displacement. Thus $f \in L^{4/3}(\mathbb{T}) \cap L^1(\mathbb{T})$. With normalized circle measure, the constrained product integral is $f^{*N}(a_\sigma)$. Indeed, $g = f * f$ belongs to L^2 by Young's inequality, and $g * g$ is bounded and continuous by Cauchy–Schwarz and continuity of translations in L^2. Further convolution with $f \in L^1$ preserves these properties. Thus point evaluation is legitimate, and Young's inequalities give

$$f^{*N}(a_\sigma) \leq \|f\|_{4/3}^4 \|f\|_1^{N-4} \leq C^N, \qquad N \geq 4.$$

The Jacobian costs C^N by Lemma 5.1. The remaining product kernel costs CN. Division by $N! \geq (N/e)^N$ bounds the integral on the disk by $N^2(C/\sqrt{N})^N$. Cauchy's inequality proves the assertion. □

6 A regular coordinate for the one-sided contour

The support of L_N can be lifted to one angular interval $(-\pi-\delta,\delta)$: the remaining upper plateau has zero weight because $p(\theta_i)=1$ would imply $P>1/2$. The bordered entries are rational and periodic in each angle, so this lift preserves the determinant and its integral. We use $w_i=\theta_i-ih_i$ and the physical phase $\varphi_i=\arccos(S-\cos w_i)$ throughout.

Choose a fixed branch width, decreased further below if needed, so that

$$\delta_B<\frac{1}{4}\min_{\substack{3\le n\le p\\ n\ \text{odd}}}\operatorname{dist}(-b,(2\pi\mathbb{Z}+a_\sigma)/n). \tag{6.1}$$

The minimum is positive because $e^{-inb}\neq\sigma$ for every n. This finite gap is used only for the lower-order complement in Section 10; the infinite-tail estimate below does not require any mean to avoid the branch. The right interval will be $[-b-\delta_B,\delta]$; its complement on the one-sided support has a compact left root. The displacement strength is chosen after these fixed margins and cutoffs.

Use the sum coordinate

$$v=\varphi+w.$$

It is regular even at the lower inverse-dispersion branch.

Lemma 6.1 (Regular inverse and right-side convexity)**.** *The one-sided limiting dispersion curve has a fixed complex neighborhood on which v is a regular coordinate and $w=U(t,v)$ is analytic, with bounded fixed joint parameter and coordinate jets. On a fixed convex neighborhood $\mathcal{D}_R$ of the right v-interval, enlarged to contain a short negative branch segment,*

$$\operatorname{Re}U_{vv}(t,v)\ge c>0.$$

The actual right-contour images lie in $\mathcal{D}_R$ after the branch width, deformation and parameter neighborhood are fixed small.

Proof. On the limiting compact right arc put $g(\theta)=\arccos(S_*-\cos\theta)$. On $[-b,0]$, $g'(\theta)=-\sin\theta/\sin g(\theta)\ge 0$. Since $g'(0)=0$, a short upper sliver retains $1+g'(\theta)>c>0$. Thus $v=g(\theta)+\theta$ is strictly increasing. At the lower branch use φ itself: $w=-\arccos(S(t)-\cos\varphi)$ is even analytic in φ, so $\mathrm{d}(w+\varphi)/\mathrm{d}\varphi=1$ there. Thus v is regular through the branch.

There is an explicit right inverse:

$$U(t,v)=\frac{v}{2}-\arccos\left(\frac{S(t)}{2\cos(v/2)}\right). \tag{6.2}$$

Its branch is fixed by $w<\varphi$ on the limiting right arc. The real v-interval is contained strictly in $(-2\widehat{\alpha},2\widehat{\alpha})$, since $b<2\widehat{\alpha}$ and the upper sliver is short. Put $x=v/2$ and $D=4\cos^2x-S_*^2>0$. Direct differentiation gives

$$U_{vv}(t_*,v)=\frac{S_*}{4\cos^2x\sqrt{D}}+\frac{S_*\sin^2x}{D^{3/2}}>0. \tag{6.3}$$

Its branch-endpoint value is $1/\sin b$. Compactness supplies a positive lower bound, retained on a thin fixed convex rectangle and for t in a small complex neighborhood.

On the remaining compact left arc, $\varphi=-\mathrm{i}\gamma$ with $\gamma>0$, so v has real part θ and is injective. There $1+g'(w)$ cannot vanish because g' is purely imaginary. Both endpoint slivers remain regular after a small extension. The left image has real part less than $-b$, and the right image has real part greater than $-b$; they meet in the regular branch chart. An analytic inverse therefore exists along the embedded compact arc, with bounded fixed jets after shrinking its neighborhood. Choose the deformation and parameter margins inside this neighborhood. □

Let $\phi(t,v) = v - U(t,v)$ and $D(t,v) = 4\cos^2(v/2) - S(t)^2$. The square root below is fixed by $\sqrt{D} = \sin\phi - \sin U$; it is positive on the limiting right arc and continued along the one-sided inverse. Implicit differentiation of $\cos U + \cos\phi = S$ gives the exact formulas

$$U_v = \frac{\sin\phi}{\sqrt{D}}, \qquad U_t = \frac{S'}{\sqrt{D}}, \qquad a(t,v) := \frac{U_v}{\sinh(\mathrm{i}\phi)} = -\frac{\mathrm{i}}{\sqrt{D}}. \tag{6.4}$$

The denominator never vanishes on the chosen neighborhoods. These formulas directly prove that the regular one-body coefficient and its joint jets are bounded, including at the inverse branch.

The ordinary bordered kernel is analytic in these regular coordinates. Near the double branch, the even inverse in φ cancels $\varphi_i + \varphi_l$ from its denominator. Away from there the denominator is separated: the double lower branch is the only limiting pair-product zero on the one-sided support. In particular,

$$|L(v_i, v_l)| \le C|v_i - v_l| \quad \text{on } \mathcal{D}_R^2, \tag{6.5}$$

and its fixed joint jets are bounded. The diagonal zero remains in this single inverse coordinate even at the branch.

6.1 Square-root geometry on the physical contour

Lemma 6.2 (Branch normal form)**.** *Put $u = \theta + b$ and $a_B = \sin b > 0$. On the original unit contour, and on a true lower branch displaced outward by a common $\mu_0 \ge 0$, set $E_0 = \varepsilon + \mu_0$. Uniformly for the fixed small branch arc and $0 \le \mu_0 \le \tau$, one has*

$$\begin{aligned} &D = 1 - W = a_B u - i\beta + R, \qquad && cE_0 \le \beta \le CE_0, \\ &R \in \mathbb{R}, \quad |R| \le C(u^2 + E_0^2), \qquad && D_u = a_B + O(|u| + E_0). \end{aligned}$$

Here β may depend smoothly on u, with $|\beta_u| \le CE_0$. On the physical fourth-quadrant branch,

$$\varphi = \sqrt{2D}\, h(D), \qquad h(0) = 1, \qquad |\varphi'| \asymp (|u| + E_0)^{-1/2}, \tag{6.6}$$

where h is analytic. Moreover

$$-\Im\varphi \ge c\sqrt{|u| + E_0} \quad (u \le 0), \qquad \Re\varphi' \ge \tfrac{2}{3}|\varphi'| \quad (u \ge 0), \qquad \int_{|u|\le\delta_B} |\varphi'(u)|\, du \le C.$$

For coordinates with the same outward shift, a minimum nonnegative coordinate u_j satisfies $|\varphi_i'| \le C|\varphi_j'|$ for every other nonnegative coordinate.

Proof. Use $y = e^{\mu_0 + i(-b+u)}$ and the exact real and imaginary parts of $W = S - \cosh(\mu_0 + i(-b+u))$. The imaginary part is

$$\beta = \Im S + \sinh(\mu_0)\sin(b - u),$$

and $\Im S = 2\sin(\arg t_*)\varepsilon + O(\varepsilon^2)$. The real part has the asserted expansion because $1 - S_* + \cos b = 0$, $\Re S - S_* = O(\varepsilon^2)$, and $\cosh\mu_0 - 1 = O(\mu_0^2)$. The analytic factorization of $\arccos(1 - D)$ gives (6.6). The relative error with respect to $a_B u - i\beta$ is uniformly small after the branch width and τ are fixed small. Its square root lies in a small fixed enlargement of the fourth quadrant. For $u \le 0$ its imaginary part has size at least a fixed multiple of its modulus; for $u \ge 0$ the derivative lies in a cone whose angle is at most $\pi/4 + o(1)$ from the positive real axis. This proves the stated attenuation and $2/3$ cone bounds. The integrable majorant and the minimum-coordinate comparison follow immediately from the two-sided derivative estimate.

□

6.2 Coordinate size and angular diameter

Lemma 6.3. *On the actual right contour put* $u_i = \theta_i + b$, $\mu = N^{-1}\sum_i u_i$ *and* $d = \max_i \theta_i - \min_i \theta_i > 0$. *For any* r *with* $r \le d \le 2r$, *set* $T = |\mu| + r + \varepsilon$. *Then*

$$|1 + \varphi_w(w_i)| \le C(|u_i| + \varepsilon)^{-1/2}, \tag{6.7}$$

$$\int_{\mu-2r}^{\mu+2r} (|u| + \varepsilon)^{-1/2} \mathrm{d}u \le \frac{Cr}{\sqrt{T}}, \tag{6.8}$$

and

$$|U_v(t, v_i)| \le C\sqrt{T}, \qquad |v_i - v_l| + |L_{il}| \le Cr/\sqrt{T}. \tag{6.9}$$

These are pointwise estimates. No extrema, diameter, or mean scale is differentiated, and no partition of the integral is made here.

Proof. Near the lower branch, $p = 0$, $m = 1$, so the outward displacement is the same $\tau P/N$ at every such coordinate. The branch normal form gives (6.7); away from that branch compactness gives the same bound, with a larger constant. It also holds along a one-variable angular curve with the actual occupancies held fixed. The curve derivative has the additional bounded factor $1 - \mathrm{i}\tau[(P/N)m' - (Q/N)p'] = 1 + O(\tau)$.

If $|\mu| \ge 4r$, the denominator in (6.8) is comparable to $\sqrt{|\mu| + r + \varepsilon}$. Otherwise enlarge the interval to $[-6r, 6r]$ and use

$$\sqrt{6r + \varepsilon} - \sqrt{\varepsilon} = \frac{6r}{\sqrt{6r + \varepsilon} + \sqrt{\varepsilon}}.$$

This proves the interval estimate even when the interval crosses the branch. Joining two actual angles along their fixed-occupancy curve gives exactly the actual v endpoints. Integrating its derivative, and then using (6.5), proves the second inequality in (6.9).

For the remaining estimate, first suppose $P = 0$. Nonnegativity of p gives $h_i = 0$ for every angle. The branch normal form and compactness imply $|\varphi_i| \le C\sqrt{|u_i| + \varepsilon} \le C\sqrt{T}$. Equation (6.4) then gives $|U_v(t, v_i)| \le C\sqrt{T}$. If $P > 0$, some actual right angle lies in the positive support of p, hence $\max u_i \ge b$ and $T \ge b/2$. Boundedness of U_v gives the same conclusion. All estimates permit arbitrary actual occupancy fractions in $[0, 1]$.

□

7 Tangential differentiation of the complete one-sided integral

7.1 The scalar density and its full Jacobian

Let $M = D_\theta w = I - \mathrm{i}Dh$. Its inverse is uniformly bounded for small τ, $\det M$ has bound C^N, and fixed jets add polynomial factors in N. Also $\mathbf{1}^T M = \mathbf{1}^T$, hence $\mathbf{1}^T M^{-1} = \mathbf{1}^T$. For $\chi_t(\theta) = (v_i)$,

$$D_\theta \chi_t = \mathrm{diag}(1/U_{v,i})M, \quad J = \det D_\theta \chi_t = \det M \prod_i (1/U_{v,i}). \tag{7.1}$$

This identity is used at $\varepsilon > 0$, where all factors are nonzero. The apparent branch poles will be retained as integrable Jacobian factors, never differentiated as unregularized amplitudes.

The complete deformed scalar density is $Jf(t, v)$, where

$$f(t, v) = \frac{1}{N!(2\pi)^{N-1}} \frac{1 + \sigma Z}{1 - \sigma Z} \det \mathcal{A} \prod_i a(t, v_i). \tag{7.2}$$

Formula (7.2) includes the unchanged factor 2π converting the ambient normalized density to constrained Haar normalization. The entire occupancy Jacobian has cancelled from the regular amplitude.

For a residual field B in v coordinates define

$$V = -M^{-1}(\dot{U} + U_v B), \tag{7.3}$$

where a dot denotes the fixed-v derivative ∂_t and products inside the vector are componentwise. Differentiating $U(t, \chi_t(\theta)) = w(\theta)$ shows that (7.3) is exactly $(D_\theta \chi_t)^{-1}(\partial_t \chi_t - B)$. If

$$\sum_i U_{v,i} B_i = -\sum_i \dot{U}_i, \tag{7.4}$$

then $\sum_i V_i = 0$ on an ambient neighborhood. If in addition

$$\sum_i B_i = 0, \tag{7.5}$$

then $(\partial_t - V \cdot \nabla_\theta) \sum_i \varphi_i = 0$. Indeed $\sum \varphi_i = \sum v_i - \sum w_i$, and $\sum w_i = \sum \theta_i$ is frozen by tangency.

For every fixed smooth real angular weight ω the exact identity is

$$\partial_t(\omega J f) - \mathrm{div}_\theta(V \omega J f) = \omega J[\partial_t f + \mathrm{div}_v(Bf)] - (V \cdot \nabla_\theta \omega) J f. \tag{7.6}$$

To verify the identity, put $A_\chi = D_\theta \chi_t$. Commuting mixed derivatives of χ_t gives

$$(\mathrm{div}_v B) \circ \chi_t = \mathrm{tr}\left[A_\chi^{-1} \left(\partial_t A_\chi - \sum_i V_i \partial_i A_\chi \right) \right] - \mathrm{div}_\theta V.$$

Jacobi's formula therefore gives $J(\mathrm{div}_v B) \circ \chi_t = \partial_t J - \mathrm{div}_\theta(JV)$. The product and chain rules prove (7.6). Because $\sum V_i = 0$ identically, the intrinsic divergence on $\sum \theta_i = c$, parametrized by the first $N-1$ angles, is

$$\sum_{i<N} (\partial_i - \partial_N) V_i = \sum_i \partial_i V_i.$$

The same formula holds for V multiplied by any scalar density. Thus ambient divergence restricts to constrained divergence without an additional measure factor. Integration by parts applies to the real and imaginary parts of these complex fields. The identity concerns scalar coefficients with respect to the original real constrained measure on $\mathbb{H}_N^\sigma$. The bumps remain smooth functions of the real angles. Formal v differentiation is the commuting complex frame $(D_\theta \chi_t)^{-1} \nabla_\theta$; its coefficients and their derivatives are controlled below.

More explicitly, if $A_\chi = D_\theta \chi_t$, set

$$E_a = \sum_i (A_\chi^{-1})_{ia} \partial_{\theta_i}, \qquad E_t = \partial_t - \sum_i (A_\chi^{-1} \partial_t \chi_t)_i \partial_{\theta_i}.$$

Then $E_a \chi_b = \delta_{ab}$ and $E_t \chi_b = 0$. Every commutator $[E_a, E_b]$ or $[E_t, E_a]$ is a linear combination of the real angular derivatives and annihilates all χ_b. Invertibility of A_χ therefore makes every commutator zero. This is a pointwise differential-operator calculation on the original real domain, with the occupancies treated as smooth real-angle functions.

7.2 A bounded transport when a left root is present

Choose a single smooth right cutoff $r(\theta)$, equal to one for $\theta + b \geq -\delta_B/2$ and zero for $\theta + b \leq -\delta_B$, with its harmless periodic return inside the excluded upper plateau. Decompose the complete one-sided integral by

$$1 = \prod_i r(\theta_i) + \left(1 - \prod_i r(\theta_i)\right).$$

Telescope the second term. Each of its at most N terms has one fixed named anchor a in a compact left region, including on every cutoff jet. That anchor has $|z_a| \leq v_0 < 1$, and $|U_{v,a}| \geq c > 0$.

On such a term take

$$B_a = -\frac{\sum_i \dot{U}_i}{U_{v,a}}, \qquad B_i = 0 \ (i \neq a). \tag{7.7}$$

Equation (7.4) holds; equation (7.5) is unnecessary because $|1 - \sigma Z| \geq 1 - v_0$. All fixed regular-coordinate and parameter jets of B, V and the regular amplitude have polynomial N bounds, apart from fixed one-body products costing C_j^N. In particular the derivatives of $Z = \exp(-\mathrm{i}\sum \phi_i)$ are bounded by a polynomial in N, since $|Z| \leq 1$ at the actual contour and fixed jets of each ϕ_i are bounded.

Hadamard applied to the complementary minors after fixed derivatives therefore bounds the transported density by

$$C_j^N N^{-N/2+C_j} \prod_i |1 + \varphi_w(w_i)|. \tag{7.8}$$

The named compact factor is bounded. Eliminate its angle using the constraint; every remaining factor has the independent majorant $C(1 + \mathrm{dist}(\theta, -b)^{-1/2})$. Integrating the product costs C^N. This proves a uniform summable bound for every fixed derivative of the whole left-root part, including all its mixed branch configurations.

7.3 One collective field on the entire right part

Define the mixed covariance and angular sums

$$R = \sum_{i<l} (U_i - U_l)(U_{v,i} - U_{v,l}), \qquad G_i = \sum_l (U_i - U_l), \qquad A = -\sum_i U_{t,i}.$$

Off full equality define the field by

$$B_i = \frac{AG_i}{R}. \tag{7.9}$$

The denominator is nonzero on the actual right contour, as proved below. Pairing indices gives the exact identities

$$\sum_i G_i = 0, \quad \sum_i U_{v,i} G_i = R, \qquad \sum_i B_i = 0, \quad \sum_i U_{v,i} B_i = -\sum_i U_{t,i}. \tag{7.10}$$

Thus the same field freezes both required sums on the entire right part. There are no divided pair differences in its formula.

Lemma 7.1 (Protected mixed covariance)**.** *On an actual right-contour diameter shell $r \leq d \leq 2r$, put $T = |\mu| + r + \varepsilon$ and $\Delta = r/\sqrt{T}$. For fixed sufficiently small geometric choices,*

$$\mathrm{Re}(e^{-\mathrm{i}\pi/4} R) \geq \frac{c}{\sqrt{T}} \sum_{i<l} (\theta_i - \theta_l)^2 \geq c\sqrt{T}\Delta^2. \tag{7.11}$$

In particular the field is regular at every partial diagonal.

Proof. Hold the actual occupancies $\rho = P/N$, $\upsilon = Q/N$ fixed and join any two angles along

$$w(x) = x - \mathrm{i}\tau\{\rho m(x) - \upsilon p(x)\}, \qquad v(x) = \varphi_t(w(x)) + w(x).$$

This auxiliary curve has exactly the actual endpoints and $w'(x) = 1 + O(\tau)$. Its images lie in the fixed inverse domain for every $0 \le \rho, \upsilon \le 1$ by the same small-displacement bounds used in Lemma 6.1.

The positive real U_{vv} on the limiting right interval has arbitrarily small argument in its sufficiently thin neighborhood. Near the lower branch, (6.4) and the fourth-quadrant normal form put U_v in an arbitrarily small enlargement of the argument interval $[-\pi/2, 0]$. On the remaining compact right arc U_v is close to a positive real number. Consequently

$$\frac{\mathrm{d}}{\mathrm{d}x} U_v(t, v(x)) = \frac{U_{vv}}{U_v} w'(x)$$

lies in a fixed small enlargement of the cone $[0, \pi/2]$. Choose the branch width, neighborhood and τ so that the enlarged cone still has width less than π with a fixed margin. For $\theta_i > \theta_l$, its integral has the same cone, while $U_i - U_l = w(\theta_i) - w(\theta_l)$ has argument $O(\tau)$ and modulus comparable to $\theta_i - \theta_l$. Every product in R therefore has a uniformly positive $\pi/4$ projection.

Quantitatively, $|U_v| \le C\sqrt{T}$ along each joining curve. To see this, its v-oscillation is at most $Cr/\sqrt{T} = C\Delta$ by the fixed-occupancy interval estimate in Lemma 6.3. The endpoint bound $|U_v(t, v_i)| \le C\sqrt{T}$, bounded U_{vv} on the convex domain, and $\Delta \le \sqrt{T}$ give the assertion throughout the curve. Since $|U_{vv}| \ge c$ and $|w'| \ge c$, integration in the cone gives

$$|U_{v,i} - U_{v,l}| \ge c|\theta_i - \theta_l|/\sqrt{T}.$$

The positive projection of each covariance product is thus at least $c(\theta_i - \theta_l)^2/\sqrt{T}$. Summing proves (7.11), including at all partial collisions. □

Lemma 7.2 (Joint jets at the regular-coordinate scale)**.** *For every fixed total order $m = a + |\beta|$, on the same actual contour,*

$$|\partial_t^a \partial_v^\beta B_i| + |\partial_t^a \partial_v^\beta V_i| \le C_j N^{C_j} \Delta^{-1-m}. \tag{7.12}$$

Only pointwise scale envelopes are used; no mean, diameter or scale is differentiated. There is no separate inverse-ε loss.

Proof. Write $D_{il} = U_i - U_l$ and $E_{il} = U_{v,i} - U_{v,l}$. The actual contour and Lemma 6.3 give

$$D_{il} = O(\sqrt{T}\Delta), \quad U_{v,i} = O(\sqrt{T}), \quad E_{il} = O(\Delta).$$

Every positive fixed-v parameter derivative of either difference D_{il} or E_{il} is $O(\Delta)$, by bounded analytic divided differences on the convex inverse domain. Their other fixed joint jets are bounded; a first v derivative of D_{il} has the better size $O(\sqrt{T})$.

Leibniz' rule yields, for any joint derivative $\mathcal{D}^m$,

$$\begin{aligned} &|\mathcal{D}^m R| \le C_j N^{C_j} \sqrt{T}\Delta^{2-m}, \quad |\mathcal{D}^m R^{-1}| \le C_j N^{C_j} T^{-1/2} \Delta^{-2-m}, \\ &|\mathcal{D}^m G_i| \le C_j N^{C_j} \sqrt{T}\Delta^{1-m}. \end{aligned} \tag{7.13}$$

Here are the complete size checks. The R jets at orders $0, 1, 2$ are bounded by $\sqrt{T}\Delta^2$, $\sqrt{T}\Delta$, and $\sqrt{T}$; higher orders are bounded. At order one the possibly different parameter term has size $O(\Delta^2)$, which is at most $O(\sqrt{T}\Delta)$. At order two the possible sizes are $\sqrt{T}$, Δ, and Δ^2, all bounded by $C\sqrt{T}$. For G_i the corresponding sizes at orders $0, 1$ are $\sqrt{T}\Delta$, $\sqrt{T}$, and higher jets are bounded. Since $\Delta \le \sqrt{T}$ and maximum scales are fixed, these tables imply the displayed

envelopes at every fixed order. The quotient rule and (7.11) give the inverse bound. All jets of A are polynomially bounded in N. Formula (7.9) now proves the estimate for B.

The smooth-frame coefficients are exactly

$$E_a\theta_i = (M^{-1})_{ia}U_{v,a}, \qquad E_t\theta_i = (M^{-1}U_t)_i.$$

Their fixed joint jets have polynomial N bounds: differentiate M^{-1} using $D(M^{-1}) = -M^{-1}(DM)M^{-1}$ and induct on the bounded real angular derivatives of M and analytic jets of U. No U_v^{-1} enters. This also bounds every smooth cutoff jet. Formula (7.3) and the product rule give the same estimate for V. □

At fixed ε, N, the complete product identity gives collision degree $N(N-1)$ in the angular differences. The field has one inverse diameter and each transport loses at most two degrees. The full-equality puncture flux therefore vanishes through order $j \le k$ whenever

$$N^2 - 2j - 3 \ge 4N_* + 2 > 0, \qquad N \ge N_* + 2. \tag{7.14}$$

No partial-diagonal flux occurs, since Lemma 7.1 separates the sole field denominator at every nonconstant configuration. The transports are performed on the entire smooth right term before any diameter division.

There are two useful bounds for the resulting density. First, the complete pair product has the exact regular factorization

$$P_{il} = (v_i - v_l)^2 A_{il}(t, v_i, v_l),$$

where A_{il} and its fixed jets are analytic and bounded on $\mathcal{D}_R^2$. This follows from (2.1) and the analytic diagonal zero in (6.5), including at the double branch. Thus its $N(N-1)$ collision degrees remain available. A v derivative loses at most one degree, its field coefficient costs one, and a divergence costs two. Fixed-v parameter differentiation preserves the pair zeros and costs at most one scale power in previously formed field coefficients. The product rule, (7.12), and induction on the transports give

$$C_j^{N^2}\Delta^{N(N-1)-2j}\frac{\prod_i |1+\varphi_w(w_i)|}{|1-\sigma Z|}. \tag{7.15}$$

There are only polynomially many index choices at fixed j. The bounded analytic factors in all pairs cost $C_j^{N^2}$; one-body factors, cutoff jets, and the Jacobian $\det M$ cost at most $C_j^N N^{C_j}$ and are absorbed in the same constant. The factor $1/N!$ can only improve this bound. The simple kernel is retained because the field freezes $\sum\varphi_i$ exactly.

Second, on any region $d \ge d_0 > 0$, all field jets are bounded with polynomial N costs. Differentiate the bordered determinant and apply Lemma 4.4 to its complementary ordinary minors. Division by $N! \ge (N/e)^N$ gives

$$C_{j,d_0}^N N^{-N/2+C_j}\frac{\prod_i |1+\varphi_w(w_i)|}{|1-\sigma Z|}. \tag{7.16}$$

Both estimates apply to the same transported density; no cutoff between them has been differentiated.

8 One coarea estimate for the complete right integral

8.1 A projected periodic kernel

Lemma 8.1 (Oblique periodic kernel)**.** *Fix $|\vartheta| < \pi/2$ and put $P_\vartheta = 2\pi\cos\vartheta$. If $G = x - ia$, $a \ge a_0$, and $0 < a_0 \le 1$, then, with $\nu = \Re(e^{-i\vartheta}G)$,*

$$\frac{1}{|1-e^{-iG}|} \le \frac{C_\vartheta}{a_0 + \operatorname{dist}(\nu, P_\vartheta\mathbb{Z})}.$$

Consequently, on any injective ν-parametrized arc whose ν-image has length L, the integral of this majorant in ν is at most $C_\vartheta(1+L)(1+|\log a_0|)$.

Proof. For $0 < a \le 1$, use

$$|1-e^{-a-ix}|^2 = (1-e^{-a})^2 + 2e^{-a}(1-\cos x) \ge c\{a^2 + \operatorname{dist}(x, 2\pi\mathbb{Z})^2\}.$$

It follows that $|1-e^{-iG}| \ge c\{a+\operatorname{dist}(x,2\pi\mathbb{Z})\}$. For an integer n minimizing the latter distance,

$$|\nu - 2\pi n\cos\vartheta| \le \cos\vartheta\,|x-2\pi n| + |\sin\vartheta| a.$$

Thus $a_0 + \operatorname{dist}(\nu, P_\vartheta\mathbb{Z})$ is bounded by a constant times the preceding lower bound. For $a \ge 1$, $|1-e^{-iG}| \ge 1-e^{-1}$ and the proposed denominator is bounded above by $1+P_\vartheta/2$. Integration over translates of one period proves the final claim. □

For $\sigma = -1$, replace G by $G-\pi$ in this lemma; $1+e^{-\mathrm{i}G} = 1-e^{-\mathrm{i}(G-\pi)}$. Only a fixed translate of the projected period lattice changes. The derivative, interval length, and all bounds remain the same.

For the actual contours the attenuation floor is $-\log|Z| \ge cN\varepsilon$. Indeed $\Im W_i \ge c\varepsilon$, and $W_i = \cos\varphi_i$ remains in a bounded set. Writing $\varphi_i = x_i - ia_i$ gives $\Im W_i = \sin x_i \sinh a_i \le Ca_i$, hence $a_i \ge c\varepsilon$. Thus Lemma 8.1 costs at most $C(1+L)(1+|\log\varepsilon|)$ on an injective projected phase interval of length L.

8.2 Oblique extreme coarea with the full occupancy dependence

Let $I = [-b-\delta_B, \delta_R]$, where $\delta_R = \delta < b/2$, and choose the negative segment inside the lower branch plateau. The upper inverse branch is outside I. Use the original balanced contour with reflected disjoint bumps and put

$$\rho = P/N, \quad \upsilon = Q/N, \quad f(u;\rho,\upsilon) = -b+u - i\tau\{\rho m(-b+u) - \upsilon p(-b+u)\}.$$

Here $u \in [-\delta_B, b+\delta_R]$ and the original one-sided occupancies satisfy $P \le 1/2$, $0 \le Q \le N$. The auxiliary analytic estimates in terms of $E = \varepsilon + \tau\rho$ hold on the fixed box $0 \le \rho, \upsilon \le 1$. The final bound in Lemma 8.2 concerns actual occupancies computed from the angles, including on the enlarged rectangle used below. Let $\varphi_t(w) = \arccos(S(t) - \cos w)$ on the physical sheet, $b_t(w) = \partial_w\varphi_t(w)$. Use the regular coordinate defined above. The regular phase coordinate is $v_t(w) = \varphi_t(w) + w$. Its analytic inverse and the scalar transport were proved in Sections 6–7.

Lemma 8.2 (Occupancy-dependent extreme derivative)**.** *Write $G(\theta) = \sum_i \varphi_t(w_i(\theta))$ and $\vartheta = 3\pi/8$. On an actual extreme-index cell with $0 \le m = \min_i u_i < M = \max_i u_i$, all other u_i fixed, and $m+M$ fixed by the constraint, one has*

$$\Re\left(e^{-i\vartheta}\frac{dG}{dm}\right) > 0, \qquad \frac{|(b_t(w_m)+1)(b_t(w_M)+1)|}{\Re(e^{-i\vartheta}dG/dm)} \le C\frac{\sqrt{M+\varepsilon}}{M-m}. \tag{8.1}$$

The constants are independent of N, ε, and the actual occupancies. A sufficiently small fixed τ is chosen after the bump derivatives and fixed angular margins.

Proof. Temporarily hold the actual values of ρ, υ fixed and set $E = \varepsilon + \tau\rho$. On the plateau $f = -b+u-i\tau\rho$; the square-root normal form gives

$$|b_t(f(u))| \le C(u+E)^{-1/2}, \qquad |\varphi_t''(f(u))| \asymp (u+E)^{-3/2}, \qquad u \ge 0. \tag{8.2}$$

These estimates extend to the remainder of the fixed right interval by compactness: it stays a fixed distance from both dispersion branches, while u is bounded above and below there. The existing inverse supplies the curvature identity

$$-\varphi_t''(w) = \frac{U_{vv}(t,v)}{U_v(t,v)^3}, \qquad v = \varphi_t(w) + w,$$

obtained by differentiating $\varphi_w = U_v^{-1} - 1$. On the true branch with $u \geq 0$, the normal form puts U_v in an arbitrarily small enlargement of $[-\pi/4, 0]$ in argument. On the compact remainder U_v is close to a positive real number. The argument of U_{vv} is uniformly small, as in Lemma 7.1. Hence, for $\vartheta = 3\pi/8$,

$$\Re\{e^{-\mathrm{i}\vartheta}[-\varphi_t''(f(u))]\} \geq c|\varphi_t''(f(u))|. \tag{8.3}$$

Indeed the curvature arguments lie in a small enlargement of $[0, 3\pi/4]$, whose midpoint is $3\pi/8$. The same identity and the branch size of U_v give the second bound in (8.2) directly.

Put $a(u) = \partial_u f(u)$ and $B(u) = b_t(f(u))a(u)$ at these fixed occupancies. Since $a = 1 + O(\tau)$, $a' = O(\tau)$, and $|b_t(f)| \leq C|\varphi_t''(f)|$ by (8.2),

$$-B' = -\varphi_t''(f)a^2 - b_t(f)a'$$

retains (8.3). Therefore, with $d = M - m$,

$$\begin{aligned} &\Re\{e^{-i\vartheta}[B(m) - B(M)]\} \geq cL, \\ &L = \int_m^M (u+E)^{-3/2}du = \frac{2d}{\sqrt{m+E}\sqrt{M+E}(\sqrt{m+E} + \sqrt{M+E})}. \end{aligned} \tag{8.4}$$

We now restore the entire occupancy dependence rather than moving the occupancies along that auxiliary comparison curve. Set

$$B_m^{\mathrm{occ}} = \sum_v m(\theta_v)b_t(w_v), \qquad B_p^{\mathrm{occ}} = \sum_v p(\theta_v)b_t(w_v).$$

Direct differentiation of the original balanced contour gives the exact formula

$$\partial_{\theta_i} G = b_t(w_i)a(u_i) - \frac{i\tau p'(\theta_i)}{N} B_m^{\mathrm{occ}} + \frac{i\tau m'(\theta_i)}{N} B_p^{\mathrm{occ}}. \tag{8.5}$$

The constraint slice differentiates in the direction $\partial_{\theta_{\min}} - \partial_{\theta_{\max}}$. Every actual $u_v \geq m$, and the true branches have the same outward displacement $\tau\rho$. Hence (8.2) implies

$$\frac{|B_m^{\mathrm{occ}}| + |B_p^{\mathrm{occ}}|}{N} \leq C(m+E)^{-1/2}.$$

Both bump-derivative differences in (8.5) are bounded by Cd. The total occupancy correction to $B(m) - B(M)$ is thus at most $C\tau d(m+E)^{-1/2}$. Since $M + E$ lies in a fixed bounded interval, the exact expression in (8.4) gives $d(m+E)^{-1/2} \leq CL$. Choosing τ small absorbs the correction into half the positive projection in (8.4). This comparison uses the actual extreme configuration only; it does not presume a Hessian bound along a transposition path on which unchanged indices might become more extreme.

The numerator in (8.1) is at most $C[(m+E)(M+E)]^{-1/2}$, with the constant enlarged on the bounded interval. Combining with (8.4) bounds the ratio by $C\sqrt{M+E}/d$. If $P = 0$, then $E = \varepsilon$. If $P > 0$, an actual angle belongs to the positive support of p, so $M \geq b$. Therefore $E \leq C(M + \varepsilon)$ in either event, proving (8.1). □

The geometric integration domain. After all differentiation, bound the smooth weight and its transported coefficients by their proved envelopes and enlarge the nonnegative integral to the full right angular rectangle $I^N \cap \mathbb{H}_N^\sigma$. All actual occupancies there still satisfy $0 \leq \rho, \upsilon \leq 1$, so the preceding lemma applies. The balanced displacement has the same sign on each semicircle, and hence $\Im(S - \cos w_i) \geq \Im S$: the physical sheet and attenuation floor are preserved on this enlarged domain. The regular inverse also remains valid there. On the compact right arc its v-image changes by $O(\varepsilon + \tau)$; near the branch, uniform square-root

continuity bounds the change by $O(\sqrt{\varepsilon+\tau})$. Choose δ_B, τ and ε_0 after the fixed margin of $\mathcal{D}_R$ so that all these images lie inside it.

For fixed other coordinates and a fixed constrained sum, the conditions specifying the extreme indices, a diameter shell, and a nonnegative minimum are linear interval conditions in the remaining variable m, with $M = A - m$. Their intersection is therefore one interval. The negative part is bounded directly below. No assertion about the number of components of a smooth-cutoff fiber is required.

8.3 A fixed mean scale for every diameter

Write $u_i = \theta_i + b$, $\mu = N^{-1}\sum_i u_i$, and choose actual minimum m and maximum M only in the final absolute integral. The angular constraint fixes $\mu = b + (2\pi l + a_\sigma)/N$; at most CN such means meet the right interval. The independent coordinate measure after eliminating any one angle is unchanged.

On a shell $r \le d \le 2r$ use the scale $T = |\mu| + r + \varepsilon$ from Lemma 6.3. Every coordinate belongs to $[\mu - 2r, \mu + 2r]$.

For a nonnegative minimum, Lemma 8.2 gives the monotone projected phase and

$$\frac{|1+\varphi_w(w_m)|\,|1+\varphi_w(w_M)|}{|d\nu/dm|} \le C\frac{\sqrt{M+\varepsilon}}{M-m}. \tag{8.6}$$

Its phase range is at most CN, since each phase is bounded. Lemma 8.1 therefore costs $CN(H+1)$. Here and below $H = |\log\varepsilon|$.

For a nonnegative minimum, (8.6) therefore bounds the integral in the remaining extreme coordinate by

$$\int \frac{|1+\varphi_w(w_m)|\,|1+\varphi_w(w_M)|}{|1-\sigma Z|}\mathrm{d}m \le CN\frac{\sqrt{T}}{r}(H+1). \tag{8.7}$$

For a negative minimum, m lies in the fixed true-branch segment, where its attenuation gives

$$\frac{|1+\varphi_w(w_m)|\,|1+\varphi_w(w_M)|}{|1-\sigma Z|} \le \frac{C}{(|m|+\varepsilon)\sqrt{|M|+\varepsilon}}.$$

Set $a = |m| + \varepsilon$, $b_1 = |M| + \varepsilon$. Since $a + b_1 \ge M - m \ge r$, the elementary inequality

$$\frac{1}{a\sqrt{b_1}} \le \frac{1}{\sqrt{r}}\left(\frac{1}{a} + \frac{1}{b_1}\right) \tag{8.8}$$

follows by multiplying through by $a\sqrt{b_1}$ and using $a + b_1 \ge r$. On the constrained slice $M = A - m$, so each reciprocal in (8.8) integrates to $C(H+1)$ over a bounded interval. This gives

$$\int \frac{|1+\varphi_w(w_m)|\,|1+\varphi_w(w_M)|}{|1-\sigma Z|}\mathrm{d}m \le Cr^{-1/2}(H+1) \le C\frac{\sqrt{T}}{r}(H+1), \tag{8.9}$$

which is the same pair envelope as (8.7).

The two actual extreme Jacobian factors have been retained through (8.7); only the $N-2$ remaining factors are now replaced by independent one-body majorants. Their interval integrations cost $(Cr/\sqrt{T})^{N-2}$. Thus the full coarea and interval cost is

$$C^N N^C \Delta^{N-3}(H+1), \qquad \Delta = r/\sqrt{T}.$$

This includes the polynomial number of extreme-index cells and means. Combined with (7.15), the complete shell is bounded by

$$C_j^{N^2}\Delta^{N^2-2j-3}(H+1). \tag{8.10}$$

For every tail order $N \geq N_* + 2$ and $j \leq k$, the exponent is at least $4N_* + 2 > 0$.

Choose a fixed $\delta_0 > 0$ after all constants for $j \leq k$, small enough that $C_j\delta_0 \leq 1/2$ for each of them. On shells with $\Delta \leq \delta_0$, the dyadic sum converges uniformly: for fixed μ, ε,

$$\frac{\Delta(r/2)}{\Delta(r)} = \frac{1}{2}\sqrt{\frac{|\mu| + r + \varepsilon}{|\mu| + r/2 + \varepsilon}} \in [1/2, 1/\sqrt{2}].$$

Its largest term is bounded by

$$C_j^{N^2}\delta_0^{N^2-2j-3} \leq \delta_0^{-2k-3}2^{-N^2},$$

which is summable over all odd tail orders.

On the complementary shells, $\Delta \geq \delta_0$ and $T \geq r$ imply $r \geq \delta_0^2$. There are only finitely many such dyadic angular shells, uniformly in N, ε. Use (7.16); the same coarea and interval integration is at most $C_j^N N^{C_j}(H+1)$, since $\delta_0 \leq \Delta \leq C$. This gives the summable envelope

$$C_j^N N^{-N/2+C_j}(H+1).$$

Both parts of this right-side sum include every constrained mean and every odd tail order $N \geq N_* + 2$.

9 The complete logarithmic tail

Theorem 9.1. *At each positive-geometry signed point and for every $0 \leq j \leq k$, uniformly for $0 < \varepsilon < \varepsilon_0$,*

$$\sum_{\substack{N > N_* \\ N \text{ odd}}} |\mathcal{C}_N^{(j)}((1+\varepsilon)t_*)| \leq C_j(1 + |\log\varepsilon|). \tag{9.1}$$

In particular the entire higher-particle contribution is $o(\varepsilon^{-1})$ at the derivative order of the first singularity.

Proof. The selected part has the summable bound proved in Lemma 5.2, and the left-root part has the bound (7.8). For the entire right part, (8.10) and the following small/large Δ division give a summable majorant

$$\left(C_j 2^{-N^2} + C_j^N N^{-N/2+C_j}\right)(H+1).$$

These estimates cover the exact identity $\mathcal{C}_N = F_N + 2L_N$. Summing proves the theorem, and $\varepsilon(1 + |\log\varepsilon|) \to 0$ gives the required noncancellation estimate.

The choices have the following order: the selected point and k; the endpoint slivers and fixed inverse-coordinate neighborhood; the branch width, smooth bumps and right cutoff; a sufficiently small fixed τ; the parameter-disk constants; and then ε_0 and the fixed estimate threshold δ_0. The branch width may also meet the finite gap required only for Section 10. All choices are independent of the tail order N. The division by Δ is made only after the complete weighted integrals have been transported, so creates no extra derivative or boundary term. □

10 The first singularity on the constrained angular torus

Fix a prime-family point $\zeta = e^{i\alpha}$, $0 < \alpha < \pi$, $N = p$ and $k = (p^2-1)/2$. We first justify its finite-order complement using the same signed positive-geometry construction as for the tail.

Lemma 10.1 (Finite-order complement)**.** *Every fixed derivative of every odd coefficient $n < p$ is $O(1 + |\log\varepsilon|)$ along $(1+\varepsilon)\zeta$. At $n = p$, remove two small fixed reflected angular charts around $\theta_i = -\alpha$ and $\theta_i = +\alpha$ for all i. Every fixed derivative of the remaining complete coefficient is $O(1 + |\log\varepsilon|)$ on this radial approach.*

Proof. The case $n = 1$ follows from $C_1 = 1/(S-2)$. For negative S_* use (3.1); thus it suffices to work at positive S_* with the appropriate constraint σ and angle $\widehat{\alpha} \in (0, \pi/2)$. The signed $n = 1$ formula is $C_1^- = 1/(S+2)$. All remaining orders satisfy $n \geq 3$.

The finite mean gap in (6.1) places all constrained equality means away from the branch. For each fixed n, the mixed-population part F_n has all true branches displaced by at least $\tau/(4n)$ and a fixed product damping gap. For each fixed n, compactness gives a fixed parameter disk on which its density is holomorphic and bounded. Every left-root term has bounded fixed derivatives by the anchor field of Section 7.2, with its compact angle eliminated before integrating the remaining one-body majorants.

At full equality the constrained mean is $\theta_0 = (2\pi\ell + a_\sigma)/n$. The balanced shift is exactly zero there, since $P = np(\theta_0)$ and $Q = nm(\theta_0)$. The product denominator vanishes only when

$$n \arccos(S_* - \cos\theta_0) \in 2\pi\mathbb{Z} + a_\sigma.$$

For $\sigma = 1$ this is the ordinary Nickel equation; for $\sigma = -1$ translate both angles by π and use oddness of n. The arithmetic of Section 3 excludes it at $n < p$, and at p leaves only the lower center $-\widehat{\alpha}$ on the one-sided support. All other equality neighborhoods have regular roots and a separated product denominator, so their localized integrals extend holomorphically through the point.

Remove these neighborhoods and, at p, the critical chart, chosen entirely inside $m = 1, p = 0$. There the balanced deformation is the identity, $w_L = 1$, and the right cutoff is one. On the remaining right support, including all cutoff jets, $d \geq d_0 > 0$. The collective field is therefore bounded to every fixed jet order and has no puncture flux. Transport gives a bound

$$C_{n,j} \frac{\prod_i |1 + \varphi_w(w_i)|}{|1 - \sigma Z|}.$$

Retain the two extreme-coordinate factors for the coarea estimate before bounding the others by independent majorants. There are finitely many constrained means and shells with $d \geq d_0$, so Section 8 gives $O(1 + |\log\varepsilon|)$ for every fixed derivative of this remainder. Smooth-cutoff fibers are enlarged to the same geometric intervals used there.

Reflection in the complete identity $\mathcal{C}_n = F_n + 2L_n$ accounts for the two removed charts. For the negative half, negation and conjugation map these charts back to $\pm\alpha$ and preserve the bound. □

Near the lower physical chart write $\theta_i = -\alpha + u_i$, $u_N = -\sum_{i<N} u_i$. Because $p\alpha \in 2\pi\mathbb{Z}$, this is the literal constraint with measure $\prod_{i<N} \mathrm{d}u_i/(2\pi)$. At $t_\varepsilon = (1+\varepsilon)\zeta$, the physical phase satisfies

$$\varphi_i = \alpha - 2\mathrm{i}\varepsilon + u_i - \cot\alpha\, u_i^2 + O(\varepsilon^2 + \varepsilon|u_i| + |u_i|^3).$$

Indeed $\varphi_t = -2\mathrm{i}\zeta^{-1}$, $\varphi_u = 1$ and $\varphi_{uu} = -2\cot\alpha$ at the center. For $q(t,u) = 1 - Z(t,u)$ this gives

$$q(t_\varepsilon, u) = 2p\varepsilon - \mathrm{i}d_\alpha \sum_i u_i^2 + O(\varepsilon^2 + \varepsilon|u| + |u|^3), \quad d_\alpha = \cot\alpha \neq 0, \quad q_t(\zeta, 0) = 2p\zeta^{-1} =: D_t. \tag{10.1}$$

The two leading terms have perpendicular phases. Consequently, on a sufficiently small fixed shape ball,

$$|q(t_\varepsilon, u)| \geq c(\varepsilon + |u|^2). \tag{10.2}$$

To check the remainder, bound it by $C(\varepsilon_0+\delta)(\varepsilon+|u|^2)$ for $|u|\le\delta$, $\varepsilon\le\varepsilon_0$, and choose both radii so this is less than half the leading lower bound. This argument works for either sign of d_α.

With a fixed smooth shape cutoff χ equal to one near zero, the lower chart is exactly

$$I_p(t)=\int_{\mathbb{R}^{p-1}}\frac{\chi(u)\Delta(u)^2A(t,u)}{q(t,u)}\,\mathrm{d}u_1\cdots\mathrm{d}u_{p-1}. \tag{10.3}$$

Here $\Delta(u)=\prod_{i<j}(u_i-u_j)$ and A is analytic. Its value, including all physical factors, is

$$A(\zeta,0)=-\mathrm{i}a_p,\qquad a_p=\frac{2^{1-p(p-1)}}{p!(2\pi)^{p-1}\sin^{p^2}\alpha}>0. \tag{10.4}$$

Each sine numerator gives half the angle difference, each hyperbolic denominator tends to $\mathrm{i}\sin\alpha$, each one-body factor tends to $(\mathrm{i}\sin\alpha)^{-1}$, and $1+Z\to2$. Their combined phase is $(-\mathrm{i})^p(-1)^{p(p-1)/2}=\mathrm{i}^{-p^2}=-\mathrm{i}$. Every parameter derivative retains the full Vandermonde because the real shape coordinates are fixed. The upper chart is exactly equal under reflection. There are two centers; exchanging equal dispersion angles creates no additional center.

Lemma 10.2 (A nonzero radial period)**.** *Let*

$$\mathcal{S}_p=p^{-1/2}\int_{\substack{\sum_i\omega_i=0\\|\omega|=1}}\Delta(\omega)^2\,\mathrm{d}\sigma(\omega)>0.$$

Then, with $v_p=-\sum_{i<p}v_i$,

$$J_\alpha=\int_{\mathbb{R}^{p-1}}\frac{\Delta(v)^2}{(2p-\mathrm{i}d_\alpha\sum_i v_i^2)^{k+1}}\,\mathrm{d}v_1\cdots\mathrm{d}v_{p-1}=\frac{\mathcal{S}_p}{2k(2p)|d_\alpha|^k}>0. \tag{10.5}$$

Proof. The coordinate measure on the zero-sum hyperplane is its Euclidean measure divided by $\sqrt{p}$. The Vandermonde is positive on an open spherical set after squaring, so $\mathcal{S}_p>0$. The radial exponent is $p^2-2=2k-1$. With $Q=2p$ and $c=-\mathrm{i}d_\alpha$, put $x=|v|^2$ and use

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{cx}{Q+cx}\right)^k=kQc^k\frac{x^{k-1}}{(Q+cx)^{k+1}}.$$

The bracket has endpoint values zero and one, and $Q+cx$ never vanishes for $x\ge0$. The integral is absolutely convergent with radial tail R^{-3}. Since $k\equiv0\pmod4$, $c^k=|d_\alpha|^k$. Polar coordinates now give (10.5) directly. □

Theorem 10.3 (The complete first odd singularity)**.** *At the selected prime root,*

$$C_p^{(k)}(t_\varepsilon)=\Lambda_\alpha\varepsilon^{-1}+o(\varepsilon^{-1}),\qquad\Lambda_\alpha=-2\mathrm{i}a_pk!(2p)^k\zeta^{-k}J_\alpha\neq0.$$

Proof. The complement is $O(1+|\log\varepsilon|)$ by Lemma 10.1. Differentiate (10.3) k times. The unique highest pole is

$$\int\frac{\chi(u)\Delta(u)^2(-1)^kk!q_t(t_\varepsilon,u)^kA(t_\varepsilon,u)}{q(t_\varepsilon,u)^{k+1}}\,\mathrm{d}u.$$

Scale $u=\sqrt{\varepsilon}\,v$. The exponent is $[p(p-1)+(p-1)]/2-(k+1)=-1$. Equation (10.2) gives the integrable majorant $C\Delta(v)^2/(1+|v|^2)^{k+1}$. Dominated convergence gives the single-chart coefficient $-\mathrm{i}a_pk!(2p)^k\zeta^{-k}J_\alpha$. Every lower pole power $m+1$, $m<k$, contributes $o(\varepsilon^{-1})$, since

$$\varepsilon\int_0^\delta\frac{r^{2k-1}}{(\varepsilon+r^2)^{m+1}}\,\mathrm{d}r\longrightarrow0.$$

At $m=k-1$ this is $O(\varepsilon|\log\varepsilon|)$; at smaller m it is $O(\varepsilon)$. The reflected chart gives the identical coefficient, proving the factor two. All phases and estimates hold for both signs of $\cos\alpha$. □

11 Noncancellation and the whole natural boundary

Let

$$V(t) = \sum_{\substack{N \geq 1 \\ N \text{ odd}}} C_N(t).$$

Normal convergence permits termwise differentiation in $|t| > 1$. At a selected $\zeta = e^{\mathrm{i}\alpha}$, the finitely many lower orders are logarithmically bounded by Lemma 10.1, the first term is given by Theorem 10.3, and the entire higher odd tail is logarithmically bounded by Theorem 9.1, using (3.1) on the second quadrant. Therefore

$$V^{(k)}((1+\varepsilon)\zeta) = \Lambda_\alpha \varepsilon^{-1} + o(\varepsilon^{-1}), \qquad \Lambda_\alpha \neq 0.$$

A holomorphic continuation through ζ would bound this finite derivative on a short radial segment, which is impossible. The factor Q_H is locally analytic and nonzero at every selected point. If $\mathcal{X}_H(1/t) = Q_H(t)V(t)$ continued there, division by Q_H would continue V. The same argument applies to $\chi_H(1/t) = \beta_{\mathrm{phys}}(1/t)Q_H(t)V(t)$, because the physical branch of $\beta_{\mathrm{phys}}(s) = \operatorname{arsinh}(s)/(2J)$ is locally analytic and nonzero at every selected nonaxis point.

The selected primes are dense in both upper quadrants. Complex conjugation supplies a dense obstruction in the lower quadrants. Every neighborhood of every unit-circle point, including all four axis points, contains an obstructing point. Thus no point of that circle admits local holomorphic continuation of the exterior germ. The map $s = 1/t$ is biholomorphic near every unit-circle point and gives the assertion of Theorem 1.1 for the prescribed high-temperature germ.

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